
Carnitine

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General Features

Carnitine is an amino acid made in the body that helps to facilitate the burning of fat. It primarily functions to transport long-chain fatty acids into the mitochondria of the cell, where these fats are metabolized to produce the ATP energy required to power biological activities of the cell. A good example involves the muscle contraction of the heart which is an ATP-dependent activity. The heart muscle, in particular, requires adequate concentrations of L-Carnitine due to the heart muscle's high reliance upon fat as an energy source from which it generates ATP energy for muscle contraction.¹ In the body L-Carnitine is produced in the liver, kidney and brain. Its precursor is the amino acid lysine.² Carnitine deficiency in humans was first described in 1973.³ Until then it had been assumed that individuals could synthesize adequate amounts of Carnitine, ingest adequate amounts of Carnitine from food, or meet needs by a combination of both. The discovery that some individuals require Carnitine supplementation to maintain normal energy metabolism has prompted its use in clinical trials involving patients with cardiovascular diseases, Alzheimer's disease, dementia, obesity, alcohol-induced fatty liver disease, low sperm counts and decreased sperm motility, and as an ergogenic aid to enhance performance in endurance athletes.⁴

Supplementation Studies and Clinical Application

Only L-Carnitine should be used for supplementation purposes. D-Carnitine has been shown to interfere with Carnitine metabolism and reduces fatty acid metabolism and energy production.^{5,6,7}

Also note, that when using L-Carnitine to prevent or treat dementia or Alzheimer's disease, the only form that crosses the blood-brain barrier is Acetyl-L-Carnitine.^{8,9}

1. Cardiovascular Disease

In cases of angina, congestive heart failure and other cardiomyopathies, a daily supplementation with 1,500-4,000 mg of L-Carnitine has produced significant improvement in cardiac function in many clinical trials. This effect appears to be due to improved fat burning capability by cardiac muscle with resulting increased production of ATP energy, which is the energy substrate required for heart muscle contraction.¹⁰⁻²⁶

2. Alzheimer's Disease and Age-Related Dementia

Acetyl-L-Carnitine has been shown to improve cognitive function in patients with mild to moderate Alzheimer's disease and dementia. The usual daily dosage was 1,000-2,000 mg per day. This effect appears to involve better fat burning by brain cells to produce more ATP energy, and the fact that Acetyl-L-carnitine can induce synthesis of the memory chemical, acetylcholine, and increase concentrations of the mitochondrial phospholipid known as cardiolipin, which is important for the transfer of electrons in the synthesis of ATP energy in brain cells.^{9,27-35}

3. Male Fertility

Carnitine concentrations are critical to sperm energy metabolism, affecting sperm count and sperm motility. Studies with infertile men have demonstrated that supplementation with 3,000

mg per day of L-Carnitine can significantly increase sperm counts and sperm motility. This effect is related to enhanced ATP energy production as well.^{36,37,38}

4. Endurance Performance and Fat Burning

Some preliminary trials reveal that L-Carnitine supplementation can enhance fatty acid oxidation in skeletal muscle during endurance exercise in healthy subjects and athletes. The dosage for endurance athletes is usually 2,000 mg, two to three times per day. However, this evidence requires much more substantiation. Certain obese patients have been shown to have a relative deficiency in L-Carnitine that may impair their fat-burning ability, and they may thus benefit from L-Carnitine supplementation.^{39,40,41}

5. Protection Against Drug Toxicity

L-Carnitine can protect the heart against the damaging effects of the chemotherapy drug adriamycin.⁴²

Dosage Range

For most applications, 1,500 mg to 4,000 mg is the daily dosage. For brain function, only Acetyl-L-Carnitine has been shown to be effective.

Toxicity and Contraindications

L-Carnitine is extremely safe with no significant side effects reported in any human clinical studies.

Drug-Nutrient Interactions

There are no reported drug-nutrient interactions of concern for L-Carnitine. Note that the anti-seizure drug, Valproic acid and the HIV drug, AZT (zidovudine) have been shown to deplete L-Carnitine levels in the body, and thus, supplementation with L-Carnitine may be warranted under these circumstances.^{43,44,45}

Synergistic Interactions

L-Carnitine and co-enzyme Q₁₀ appear to work synergistically, enhancing the bioenergetics of mitochondrial oxidative phosphorylation in muscle, nerve and other tissues. In turn, this increases levels of ATP energy within these tissues.⁴⁶

Choline supplementation appears to spare L-Carnitine and may increase intracellular Carnitine levels (20 mg of choline per kg of body weight).⁴⁷

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